

Appendix B - Classification Experimental Configuration

This appendix records retained implementation constants and controls for the primary classification comparison. Values are reported only when preserved by the project evidence.

Table B.1. Primary classification training and checkpoint configuration

Item	Recorded setting	Interpretation
Reference architecture	YOLO26m-cls	Common backbone for the CE reference and final proposed configuration.
Input representation	Background-removed RGB images; 224×224	Offline foreground preparation was reused across directly compared classifiers.
Partition and seed	Fixed train/validation/test manifest; seed 42	Partition membership was not regenerated per directly compared run.
Optimizer and schedule	AdamW; initial LR 0.00125; cosine schedule; final LR factor 0.01	Shared selected-model training policy.
Training budget	Maximum 30 epochs; patience 15; batch 32; four workers; mixed precision enabled	Checkpoint selected from validation-side training records.
Training augmentation	RandAugment through retained Ultralytics classification configuration	Applied to training only; exact library-defined transforms remain configuration-bound.
Reference objective	Cross-entropy	Controlled reference condition.
Final configuration	ASL-LDAM + SimAM-DCFR	ASL-LDAM is training-only; feature recalibration remains in the exported network.
Candidate preference	Deployment-oriented preference approximately $\leq 15\text{M}$ parameters	Models above this preference may remain as higher-capacity diagnostic controls but are not automatically eligible as final mobile candidates.